



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Engineering

Academic Year 2025– 2026 (Odd Semester)

Degree, Semester & Branch : V Semester B.E CSE
Course Code & Title : CCS366 & Software Testing and Automation
Name of the Faculty member : Mrs. S. Vijaya Amala Devi, AP / CSE

Innovative Practice Description

- **Unit / Topic:** II / Test Case / Learning by doing
- **Course Outcome:** CO2
- **Topic Learning Outcome:** 2b
- **Activity Chosen:** Learning by doing
- **Justification**
 - Learning by Doing is an effective educational approach that aligns well with problem-based learning, especially for topics like Test Case Design in software testing. Creating and analyzing test cases requires students to clearly understand requirements, identify inputs and outputs, and recognize potential failures. Hands-on practice helps students develop this clarity and strengthens their analytical skills.
 - To make students more familiar with designing test cases, practical exercises were provided during the class. Students were encouraged to write test cases for different scenarios, identify positive and negative test conditions, and apply techniques such as boundary value analysis and equivalence partitioning. This direct practice helped them understand how theoretical methods are applied in real software situations.
 - Students created their own test cases, executed them, and discussed the results with the class. This process made the concepts more engaging, helping students understand how good test cases uncover defects and improve software quality. Learning by Doing motivated them to explore deeper, think critically, and appreciate the importance of structured test case design.
- **Time Allotted for the Activity:** 40 Minutes
- **Details of the Implementation:**
 - The students were informed about the activity in advance, and sample test case problems were discussed beforehand so they could dedicate sufficient time to understanding the concept. This preparation helped them revise requirement analysis, input-output identification, and basic test case formats.

- The instructor explained the importance of the Learning by Doing approach and highlighted how it supports problem-based learning in software testing. Students were made aware that designing test cases requires hands-on practice to understand how software behaves under different conditions.
- The steps for designing test cases using techniques such as equivalence partitioning, boundary value analysis, and decision tables were discussed prior to the activity. This ensured students knew how to apply the correct method when writing test cases.
- A student was chosen at random to write and explain test cases for a sample problem on the board using the appropriate testing technique. This encouraged active participation and improved students' confidence in presenting their work.
- Students such as revisited the steps and collaboratively designed test cases for a newly assigned scenario. This teamwork helped reinforce their understanding and allowed them to compare different approaches to the same problem.
- Another student from the class explained the test case execution flow and demonstrated how test results help identify defects and improve software quality. The step-by-step explanation helped the entire class understand how well-structured test cases support verification.
- Students were encouraged to interact, share their ideas, and compare their test case designs with classmates. During the concluding session, the instructor discussed the final answers, clarified doubts, and highlighted common mistakes, helping students strengthen their test case writing skills.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO10	PO12	PSO1
CO1	3	3	3	1	2	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovation Practice	PO1	PO2	PO3	PO4	PO12	PSO1
	2	2	1	1	1	2
Justification for Correlation	Apply basic Knowledge and fundamentals of correlation and regression techniques for solving practical problems	Identify the need of correlation and regression techniques for developing applications	Able to design solutions for applications implementing correlation and regression techniques and find relation between data	Able to design solutions for applications implementing correlation and regression techniques and find relation between data	Ability to recognize the need for correlation and regression techniques in real world applications	Exhibit knowledge in correlation and regression techniques for software development.

- Images / Screenshot of the practice:

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CSE - A

Software Testing and Automation

Testcase for WhatsApp

1) Functional test cases

Test Case ID	Feature	Test Scenario	Test Steps	Expected Result	Status
TC01	login/signup	Verifies user can login with valid credentials	1) Open whatsapp 2) Enter registered number 3) Enter OTP	User should be logged in successfully	Pass Fail
TC02	login/signup	Verifies invalid number	1) Open whatsapp 2) Enter invalid number 3) Enter OTP	Error message are display	Pass Fail
TC03	chat	Send message to the contact	1) Open chat 2) type a message 3) Press send	Message are delivered in app	Pass Fail
TC04	chat	Send media (like image, video)	1) Open chat 2) Click attach 3) Press send	media are visible for both sender and receiver	Pass Fail
TC05	Chat	forward message	1) Long press on a message 2) click forward 3) select contact 4) send	media are visible for both sender and receiver	Pass Fail
TC06	group chat	create a group	1) New group 2) select new group 3) Add participants 4) send name	group should be created and visible in chat list	Pass Fail

TC07	Call	Make audio/ video call	1) Open contact 2) click call/ video call button	Call should connect successfully	Pass Fail
TC08	Notification	Receive message	1) Open chat 2) Receive message from another device	Notification will display in sender	Pass Fail

Figure 1: A sample test case written by the students

Reflective Critique:

- Feedback of practice from students and other stakeholders:

- ❖ Feedback of practice from students and other stakeholders:

- Students learned several concepts individually, which encouraged **self-learning** and independent thinking. They were actively involved in the session instead of relying on

one-way communication.

- The activity motivated students to **share their knowledge with peers**, helping them discuss different test case designs and understand how others approached the same problem.

❖ ***Benefit of the practice:***

- Students were able to **link ideas through discussions**, improving their ability to recall techniques such as boundary value analysis, equivalence partitioning, and positive/negative test case design more easily.
- The hands-on practice enhanced their **understanding and retention of test case preparation**, as they applied concepts directly to real scenarios.

❖ ***Challenges faced in implementation***

Slow learners initially struggled to solve test case problems due to their individual learning pace. However, they were able to understand the concepts better through peer interaction and teamwork.

References:

- ❖ <https://opentextbc.ca/teachinginadigitalage/chapter/4-4-models-for-teaching-bydoing>
- ❖ https://www.ritrjpm.ac.in/images/computerscience/33_OIT551_Learningbydoing.pdf
- ❖ <https://k12.thoughtfullearning.com/blogpost/teaching-innovation-and-problem-solving>

Signature of Faculty Member

HOD